

INDUSTRIAL TRAINING



PREMAKUMARA H.P.S.
210494D

Company & Project Overview

Consulting Engineering And Contractors (Pvt) Ltd.

Corporate Office

No.423, Nawala Road, Rajagiriya.

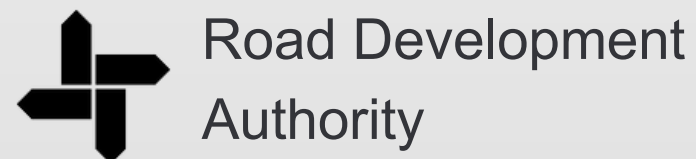
Project Office

Uduwana, Homagama.

Project : Road Construction Project

Rehabilitation, improvement, and maintenance of rural roads under the Homagama Divisional Secretariat, Colombo District, Western Province.

Contractor :



Road Development
Authority

Consultant :



Euro Group For
Engineering
Consultants



Supervisor:

Mr. G.Thanushka

Contacts : +94712493812

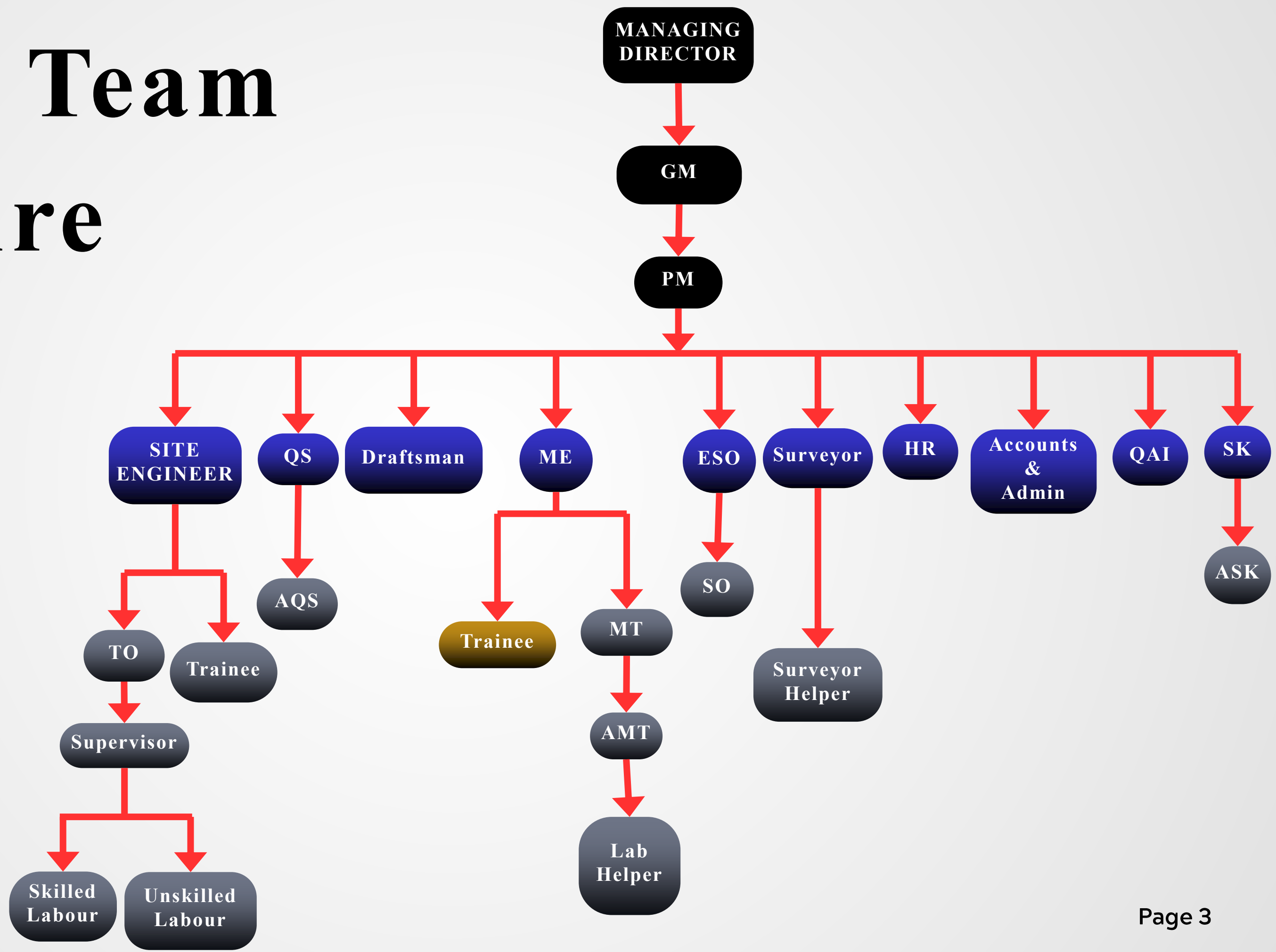
My Role : Trainee ME

Duration 24 weeks

Standard followed : ASTM (American standards) , BS (British standards)

Project Team Structure

Acronym	Meaning
GM	General Manager
PM	Project Manager
ME	Material Engineer
TO	Technical Officer
QS	Quantity Surveyor
AQS	Assistant Quantity Surveyor
ESO	Environmental & Safety Officer
SO	Site Officer
QAI	Quality Assurance Inspector
SK	Store Keeper
ASK	Assistant Store Keeper



Key Technical Learning

PO1

LO1

- **Soil & Aggregate Testing**

DCP, Proctor, CBR, PI, SA

- **Concrete Testing**

Slump, Cube test

- **Bitumen & Asphalt Testing**

Tray test, Marshall Stability, Extraction test



Design & Management

PO3

PO7

LO2

- **Designing**

Pavement design, Concrete mix design



- **Investigation & Impact**

Borrow pit investigation & environmental considerations



- **Management**

Stockpile management like ABC, SB, EMB, Aggregate and SHO and Quality testing

Documentation & QA

PO10

PO11

LO4

- Progress Reports & Inspection

DPR, RFI

- QA documentation standards

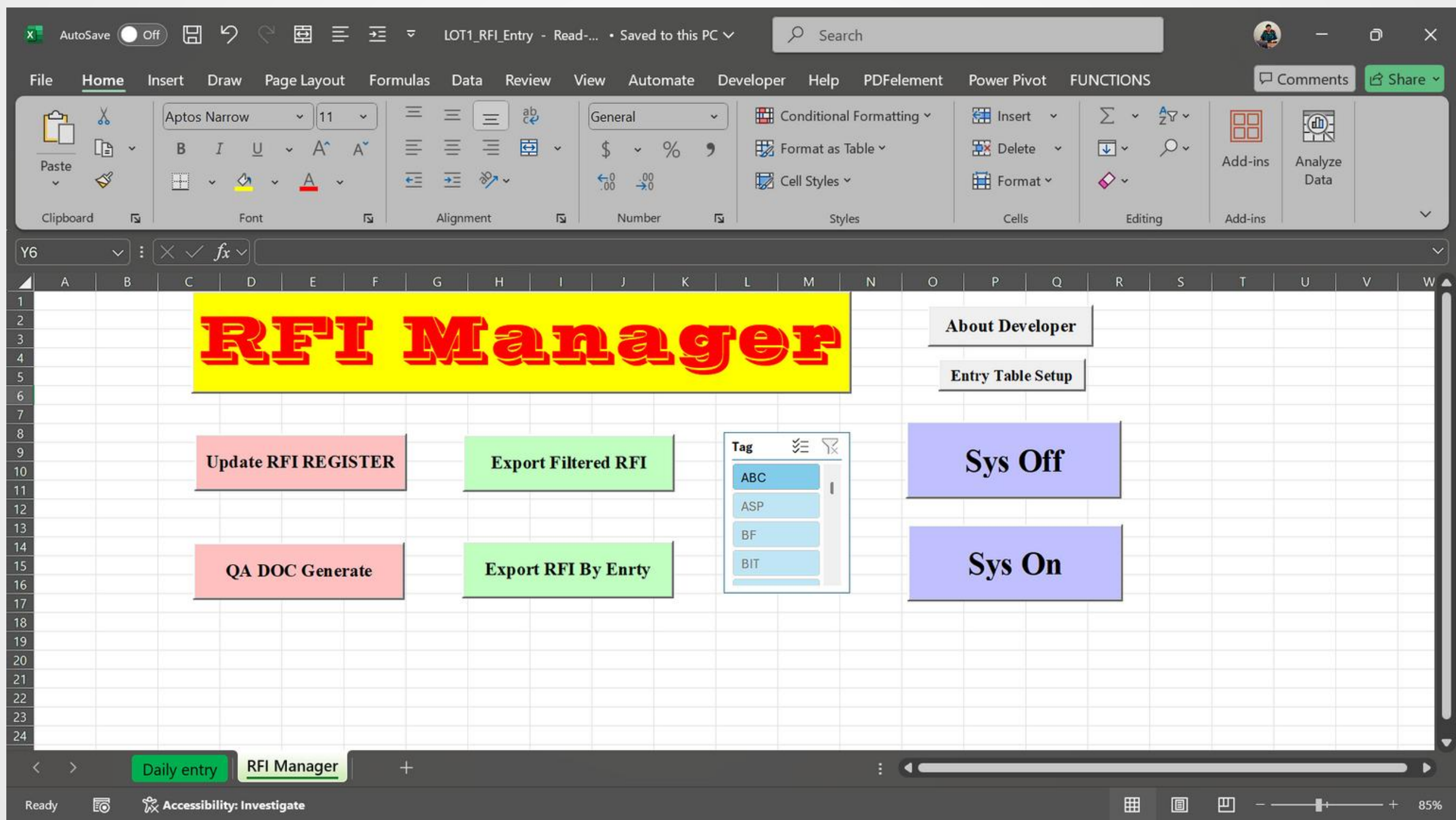
ISO 9001

- Automated reporting tools

Developed an Excel-based RFI management tool

REHABILITATION / IMPROVEMENT AND MAINTENANCE OF 8.17 KM RURAL ROADS IN HOMAGAMA DIVISIONAL SECRETARIAT IN COLOMBO DISTRICT IN WESTERN PROVINCE INCLUSIVE CONNECTIVITY AND DEVELOPMENT PROJECT-IBRD LOAN No.9290-LK (LK-RDA-418541-CW-LOT 1)	
CONSULTANT	CONTRACTOR
 EURO GROUP ENGINEERING CONSULTANTS	 CEC CONSULTING ENGINEERS & CONTRACTORS (PVT) LTD.
REQUEST FOR INSPECTION(RFI)	
Request No : CW/LOT-01/COL-15/ABC/2	RD ID : COL/15/1
Part A: Work Description (to be Completed by the Contractor)	
The following works will be ready for inspection on Date 2025-02-25 at 8:00 AM	
Work Location : 0 + 000 to 0 + 180 B/S	
Description : ABC Top Layer Laying And Compaction	
Work proposed after Inspection :	
Relevant Drawing/Sketches/Reference	Enclosure : Yes ☐ No ☐
Category of Control Point: Quality Hold Point ☐ Quality Witness point ☐ Others ☐	
From Contractor's representative	
Name : B.Y.M. Bhatiya	Signature : _____ Position : SE Date: 2025-02-25
Part B : To be completed by the Engineer	
Part B : Action	
Received by the Engineer's Representative	Date : _____ Time : _____
Inspection Required Yes ☐ No ☐	
Person Responsible for inspection	
Mr. _____ Please inspect the above work/check setting out	
Part C: Results (to be completed by the Engineer)	
The above work was inspected / Surveyed on	Sketches :
Date : _____ Time: _____	
1) Quality Notification :	
- Notice of No - Objection ☐ - Notice of No - Objection subject to comments ☐ - Works may Proceed Resubmit RFI(Please see comments below) ☐ - Notice of Rejection - Please resubmit RFI ☐	
2) Comment : _____	
Confirmed By : _____	
Name: _____ Signature: _____ Position : _____ Date : _____	
Part D : To be completed by Contractor	
Received By : _____	
Name: _____ Signature: _____ Position : _____ Date : _____	

Automated Reporting



Automated DCP test report

 PO2

 PO5

 LO4

- **Problem**

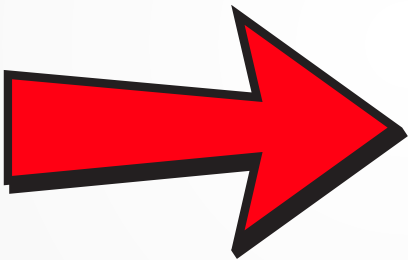
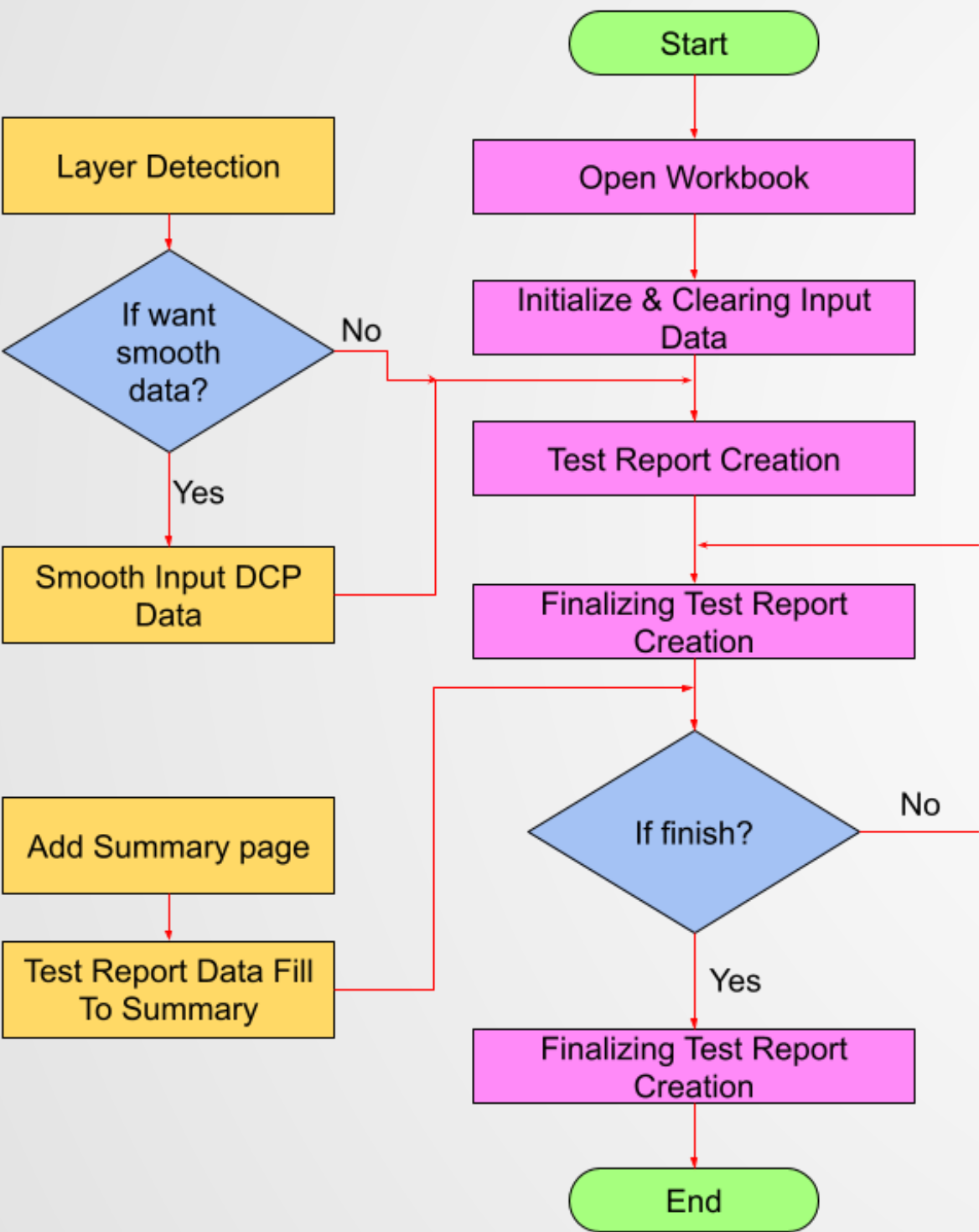
Inefficiency and errors in manual DCP test reporting.

- **Solution**

Developed an automated Excel tool for DCP test report generation.

Impact: Enhanced accuracy and efficiency, reducing management effort.

DCP test report generator



The screenshot shows the Excel interface for the DCP Test Report Generator. The 'Input' tab is active, displaying various test parameters. A large red arrow points from the flowchart to this spreadsheet. On the right side of the spreadsheet, there are three buttons: 'Create Test Report' (green), 'Finish Test Report Creation' (yellow), and 'Clear Input Data' (pink). A red arrow points to cell P2 with the text 'Insert zero reading here'.

Report No.	RDA/ICDP/LOT13/LAB/7-1/DCP/002										Blows	Penetration
Date of Test	4/18/2025										0	160
Road Name	Uduwapitiya Road										1	210
Test Location	0	+	430	LHS							1	232
Off set (m)	1.4										1	255
LOT number	3										1	278
Total road distant	17.06	Km									1	300
Contractor	Tested By	W.G.S. Lakmal								MT	1	308
	Checked By	Y.S.M Harischandra								ME	1	316
	Checked By / Witnessed By	R.M.S.K. Bandara								MT	1	325
											1	333
											1	341
											1	349
											1	357
											1	365
											1	374
											1	382

Health, Safety & Environmental Practices

PO6

LO3

- Followed proper safety protocols

During Testing, On site activities



Health, Safety & Environmental Practices

PO6

LO3

- Crack survey

- Environmental parameter monitoring

CO2 , NO2 , Weather condition



Health, Safety & Environmental Practices

PO6

LO3

- Daily Toolbox Meeting / Safety Briefing



Ethics & Life-long Learning

 PO8

 PO12

Practiced professional ethics in reporting accurate test results

Adapted to unfamiliar tools and tasks quickly

Learning beyond university curriculum

Conclusion

Program Outcomes (POs) Achieved:

- **PO1:** Engineering knowledge
- **PO2:** Problem analysis
- **PO3:** Design/development of solutions
- **PO4:** Investigation
- **PO5:** Modern tool usage
- **PO6:** Engineer and society
- **PO7:** Environment and sustainability
- **PO8:** Ethics
- **PO9:** Individual and teamwork
- **PO10:** Communication
- **PO11:** Project management and finance
- **PO12:** Life-long learning

Learning Outcomes (LOs) Achieved:

- **LO1:** Applied fundamentals of materials science and technology to industrial processes.
- **LO2:** Enhanced efficiency and suitability of industrial management practices.
- **LO3:** Identified and implemented effective Health, Safety, and Environmental (HSE) practices.
- **LO4:** Gained thorough understanding of various managerial practices within the organization.

Acknowledgment

I express my sincere gratitude to:

- Supervisors and Engineers from Consulting Engineering & Contractors (Pvt) Ltd.
- The Road Development Authority (RDA) and Euro Group Consultants for continuous guidance.
- Academic staff from the Department of Materials Science and Engineering for their invaluable support.
- Laboratory and technical staff for assistance during practical testing and site activities.

Thank You